Ishvik Kumar Singh

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EDUCATION

University of Illinois Urbana-Champaign **Jan 2025–May 2030**
Ph.D. in Electrical and Computer Engineering GPA: 3.93/4.0

Indian Institute of Technology Delhi **Jul 2018–May 2022**
Bachelor of Technology in Electrical Engineering GPA: 9.0/10

RESEARCH INTERESTS

Operating Systems, Computer Networks, and ML Systems; broadly interested in hardware–software co-design for high-performance systems.

PUBLICATIONS

Rearchitecting Storage Stacks for Host Interconnect Bottlenecks
Ishvik Kumar Singh, Shreyas Kharbanda, Rachit Agarwal, Saksham Agarwal. *Under review.*

Virtually Free: Making Strict I/O Memory Protection Feasible for Virtualized Environments
Leshna Balara, Siyuan Chai, **Ishvik Kumar Singh**, Owen Cochell, Jiyuan Zhang, Rachit Agarwal, Tianyin Xu, Saksham Agarwal. *Under review.*

RESEARCH EXPERIENCE

Rearchitecting Storage Stacks for Host Interconnect Bottlenecks **Jan 2025–Present**
Advisors: Prof. Saksham Agarwal (UIUC) & Prof. Rachit Agarwal (Cornell)

- Developed a mechanism to detect host-interconnect bottlenecks, particularly memory-interconnect contention that degrades throughput and inflates tail latency.
- Rearchitected the Linux block layer with a congestion-control mechanism that dynamically regulates in-flight I/O requests, reducing p99 latency by up to 5× and improving throughput by 15%+ under contention.

Fast and Safe I/O Memory Protection in Virtualized Environments **Jan 2025–Present**
Advisors: Prof. Saksham Agarwal (UIUC), Prof. Rachit Agarwal (Cornell) & Prof. Tianyin Xu (UIUC)

- Characterized IOMMU memory-protection overheads in virtualized systems targeting terabit-scale I/O devices.
- Redesigned the virtualization stack to achieve near-unprotected I/O performance while preserving Linux's strict I/O memory-protection guarantees.

Formalizing Safety–Performance Trade-offs in I/O Memory Protection **Jan 2026–Present**
Advisors: Prof. Saksham Agarwal (UIUC), Prof. Rachit Agarwal (Cornell) & Prof. Tianyin Xu (UIUC)

- Developing a formal event-based framework that defines safety conditions for IOMMU-based memory protection.
- Applying the framework to the Linux networking stack across kernel versions to characterize safety–performance trade-offs and identify optimizations that improve I/O performance while preserving the safety conditions.

GPU Memory Pooling within Scale-up Domains for LLM Serving **Jul 2026–Present**
Advisor: Prof. Saksham Agarwal (UIUC)

- Exploring coherent NVLink P2P access to decouple GPU compute from local HBM capacity, enabling direct remote-HBM access without explicit data movement.
- Leveraging pooled HBM capacity and bandwidth to mitigate memory fragmentation and increase LLM serving throughput under latency SLO constraints.

INDUSTRY EXPERIENCE

Intel Corporation **Jul 2022–Dec 2024**

Pre-Silicon Validation Engineer, Network and Edge (NEX) Group Hyderabad, India

- Member of emulation team; built FPGA-friendly models and validated SoC features.
- Performed pre-silicon validation for Sapphire Rapids, Granite Rapids, and Bartlett Lake across CPU microarchitecture, power management, security, cache coherence, and memory ordering.

TECHNICAL SKILLS

- **Languages:** C/C++, Python, CUDA, Verilog
- **Systems & Tools:** Linux Kernel, PyTorch, vLLM, NVIDIA Nsight, perf

RELEVANT COURSEWORK

UIUC: Modern Cloud Infrastructure, Cloud Storage Systems, High-Speed/Programmable Networks, Advanced Distributed Systems, Applied Parallel Programming.

IITD: Computer Architecture, Operating Systems, Machine Learning, Theory of Computing, Probability Theory.